

DDE Fisher Forecast and Profile-Likelihood Report

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This report summarizes the corrections made to the full P1D Fisher forecast, the DESI-based Fisher extension, and the profile-likelihood outcomes. Yes: both fiducial choices were used in the Fisher runs: (1) best-fit model vectors and (2) actual measured data vectors.

What was wrong before

- The old full Fisher used the dimensionless template (T-1) directly as the signal.
- But the covariance matrix is for P1D values in $(\text{km/s})^2$, so units were inconsistent.
- Best-fit files had 14 parameters while GP prediction expected only 11, causing shape/assert errors.

How the corrected Fisher is now computed

- Compute ratio = $T(k,z)^{-1}$ (dimensionless).
- Build physical signal $\Delta P(k,z) = P_{\text{fid}}(k,z) * \text{ratio}$ (km/s).
- Compute S/N from $SN^2 = \Delta P^T C^{-1} \Delta P$ (dimensionless and unit-consistent).
- Convert to required data multiplier with Need $N_x = (2/SN)^2$.
- Parameter fix: first 11 entries go to GP prediction; last 3 nuisance parameters are applied as data corrections.

Compatibility statement

All-z DESI Fisher ($S/N=4.7648$) and corrected eBOSS/BOSS Fisher (S/N approximately 4.71 to 4.75 depending on fiducial) are very compatible.

Table 1. Corrected Full-P1D Fisher Results (eBOSS/BOSS)

Method	Config	S/N	Need Nx
Chains best-fit fiducial	all-z	4.7468	0.18x
Data-vector fiducial	all-z	4.7115	0.18x
Data-vector fiducial	no z=2.2	4.5226	0.20x
Data-vector fiducial	no z=2.2,2.4	4.2232	0.22x

Table 2. DESI Fisher Results (DESI EDRP QMLE data vector + covariance)

Config	S/N	Need Nx
all-z	4.7648	0.18x
no z=2.2	4.4769	0.20x
no z=2.2,2.4	4.1745	0.23x

Note: Fisher runs were done with both fiducial choices where applicable:

- Best-fit fiducial vectors (from chains)
- Actual measured data vectors (data-vector fiducial)

Table 3. Profile-Likelihood Results at epsilon=1

Config	Delta chi2	sigma=sqrt(Delta chi2)	Preferred epsilon
all-z	47.2630	6.875	0.00
no z=2.2	55.7544	7.467	0.00
no z=2.2,2.4	44.2866	6.655	0.00

Result summary

- Profile scans still prefer epsilon = 0.00 in all tested configurations.
- Delta chi2(epsilon=1) is positive in each case, so full epsilon=1 is disfavored in this fixed-parameter scan setup.

Where the result files are stored

- Corrected eBOSS/BOSS Fisher: dde_analysis/results/fisher_corrected/
- Data-vector Fisher (eBOSS/BOSS): dde_analysis/results/fisher_data_vector/
- Chains-bestfit Fisher: dde_analysis/results/fisher_chains_bestfit/
- DESI Fisher (EDRP QMLE): dde_analysis/results/fisher_desi_edrp/
- Profile likelihoods: dde_analysis/results/all_zbins/, dde_analysis/results/no_z2p2/, dde_analysis/results/no_z2p2_no_z2p4/